

A method and system for determining the entropic response profile of a material is disclosed. The system receives fundamental atomic property data for a material, the data comprising at least an atomic mass and a chemical family classification. This data is processed by a predictive engine, which includes a machine learning model trained on experimental data from ultracold plasma expansion tests.

The engine generates a predicted entropic response profile for the material. This profile includes novel physical properties, namely a predicted "entropic relaxation time" (τ) and, for certain material types, a predicted "entropic buffer time" (t_{break}).

The predictive model is configured to apply a baseline scaling law, wherein the relaxation time scales with atomic mass for stable elements (e.g., noble gases), and to apply a quantifiable "entropic anomaly" based on the material's chemical family. The anomaly accounts for an accelerated relaxation for reactive elements (e.g., alkali metals) and a two-stage buffered relaxation for highly electronegative elements (e.g., halogens). The predicted profile can be used to classify the material or to select materials for specific applications requiring controlled energy release or storage characteristics.